A

Industry Apprenticeship Report

Submitted to

KAUSHALYA THE SKILL UNIVERSITY

in partial fulfillment of the requirements for the degree of

Bachelor of Computer

Application

in Cloud Computing

By

Gohel Harsh R

(1C2300204011003)

Under the Guidance

of

Dr. Parag Shukla

School of Computing

Kaushalya The Skill University

Ahmedabad - 382424

December 2025

Certificate

This is to certify that project work embodied in this report entitled Industry Apprenticeship Report as carried out by **Gohel Harsh R (Enr No:1C23002040110003)**, under my guidance in partial fulfilment of the degree of Bachelor of Computer Application in Cloud Computing, School of Computing, Kaushalya the Skill University, Ahmedabad during the academic year 2024-2025.

Date:

Place:

Guided By:

Dr. Parag Shukla
Head of the Dept.
School of Computing
KSU, Ahmedabad

Signature of Examiner

Signature of External Guide

School of Computing
Kaushalya The Skill University
Ahmedabad – 382424
December 2025

Index

1.Introduction

In today's digital era, Artificial Intelligence (AI) and Machine Learning (ML) play a crucial role in automating and improving decision-making systems across industries. High-quality and well-structured data is the foundation of any successful AI/ML model. This internship provided me with hands-on exposure to how raw, unstructured documents are transformed into meaningful datasets that can be directly used for training intelligent systems.

This report presents a detailed overview of the work performed during my industry apprenticeship at Sahana System Limited, where I worked as an AI/ML Data Annotation Intern.

The primary objective of this internship was to understand how raw and unstructured data is transformed into high-quality, machine-readable datasets used for training Artificial Intelligence and Machine Learning models.

2. Internship Role & Responsibilities

As an AI/ML Data Annotation Intern, I collaborated closely with data engineers and AI development teams. My responsibilities covered the entire dataset creation pipeline.

2.1 Downloading and Managing Raw Data

The initial stage of the project involved handling raw data provided by the company. This task included:

• Data Downloading

- Receiving PDFs through internal company portals, email, or shared cloud drives.
- Ensuring files were downloaded safely without corruption.
- Checking the total number of files and matching them with the assigned tasks.

• Data Quality Check

I was instructed to review each PDF carefully:

- Checking clarity and resolution
- Identifying blurred/unclear pages
- Ensuring proper page formatting
- Noting missing pages or inconsistencies

• Folder Structure & Organization

To maintain discipline:

- Files were grouped into project-specific folders
- Naming conventions were followed (e.g., File_001, File_002)
- Duplicate or irrelevant files were separated
- A task tracker sheet was maintained

This step ensured the entire project started with clean, organized input data.

2.2 Uploading Documents for Annotation

Before annotation, the PDFs needed to be uploaded into internal systems:

• Uploading to Company Portal

- Uploading raw PDF files to internal servers
- Updating the status in task management tools (Excel/Google Sheet/Jira)
- Monitoring successful uploads and re-uploading failed files

• Converting PDFs into Images (if required)

Some ML projects require annotation on images instead of PDF pages.
I performed:

- PDF-to-image conversion (JPG/PNG)
- Splitting multi-page PDFs into individual pages
- Ensuring every page was correctly exported and readable

Uploading is an important step because the annotation process begins only after proper file preparation.

2.3 Manual Data Annotation (Core Task)

Annotation was the main part of my internship.
I worked on complex annotation tasks that included:

• Bounding Box Annotation

Drawing rectangular boxes around:

- Text sections
- Tables
- Headings
- Fields
- Specific objects

These bounding boxes help ML models detect and identify relevant areas on a page.

• Labeling Objects and Text

Each bounding box was assigned a label depending on the project.
Example labels:

- Title
- Paragraph
- Figure
- Table
- Field Name
- Value

• Highlighting Important Information

Marking:

- Key words
- Data fields
- Document sections

• Categorization

Documents were classified based on content type, such as:

- Invoices
- Forms
- Reports
- ID documents
- Statements

• Following SOPs

I strictly followed guidelines given by the company:

- Minimum and maximum bounding box padding
- Label consistency rules

- Color-coding (if required)
- Annotating every required element

Through this, I understood how annotation accuracy directly contributes to AI model performance.

2.4 Annotation Verification

Verification is an extremely important stage.
Every annotation was checked for:

- • Label Accuracy

Is every label assigned correctly?

• Bounding Box Precision

- Is the box too tight or too wide?
- Are important parts cut off?
- Are boxes overlapping unnecessarily?

• Missing Annotations

Ensuring nothing was missed:

- Tables
- Paragraphs
- Images
- Headers or footers

• Correction of Errors

I corrected:

- Wrong labels
- Overlapping boxes
- Irregular placements
- Misclassified elements

Verification improves dataset reliability and removes annotation noise.

2.5 Using Roboflow for Annotation Automation

Roboflow is a professional dataset management platform.
I used it for:

- • Dataset Uploading
- Importing images/PDFs
- Organizing them into projects
- Creating dataset versions

• Annotating Using the Roboflow Interface

- Drawing bounding boxes
- Using label presets
- Maintaining consistency

• Model-Assisted Labeling

Roboflow supports AI-assisted annotation:

- It auto-suggests bounding boxes
- Reduces manual effort
- Speeds up annotation workflow

• Data Versioning

Each dataset update becomes a new version:

- Version 1 → Raw annotated
- Version 2 → Verified
- Version 3 → Cleaned

• Exporting Annotations

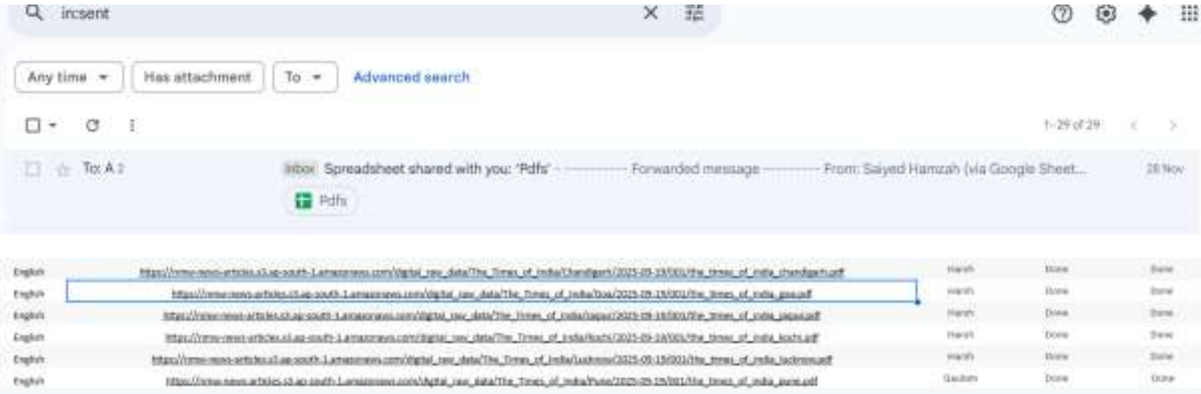
Exporting datasets to:

- YOLO format
- COCO JSON
- Pascal VOC XML
- CSV formats

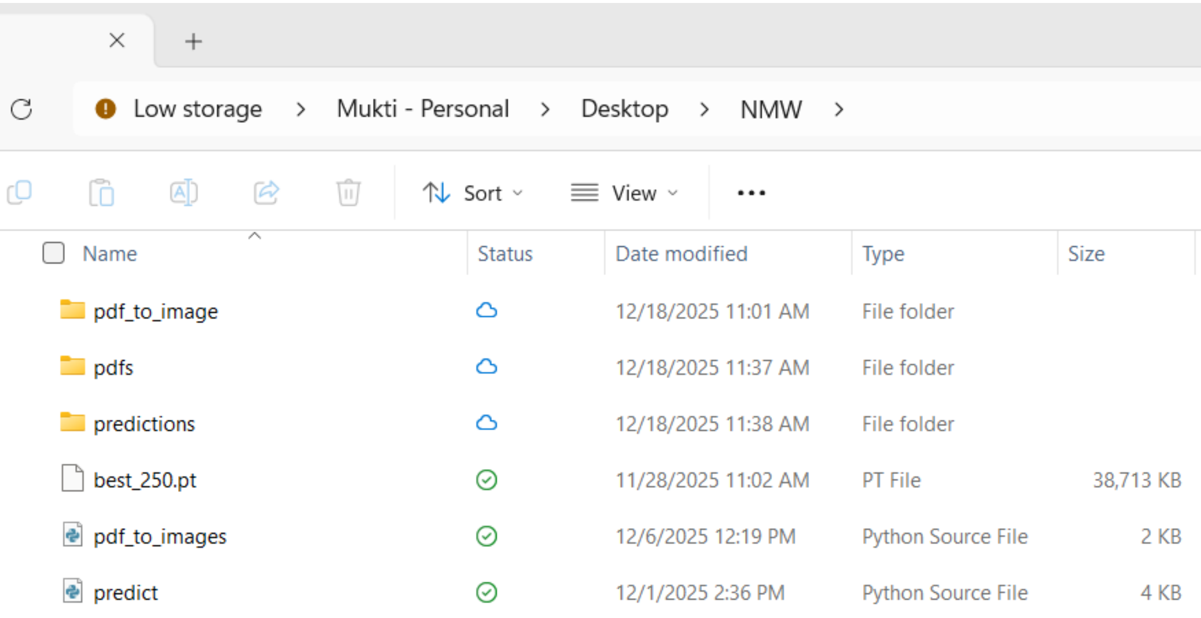
This ensures the data can be used in model training pipelines.

3. Detailed Workflow Followed During Internship

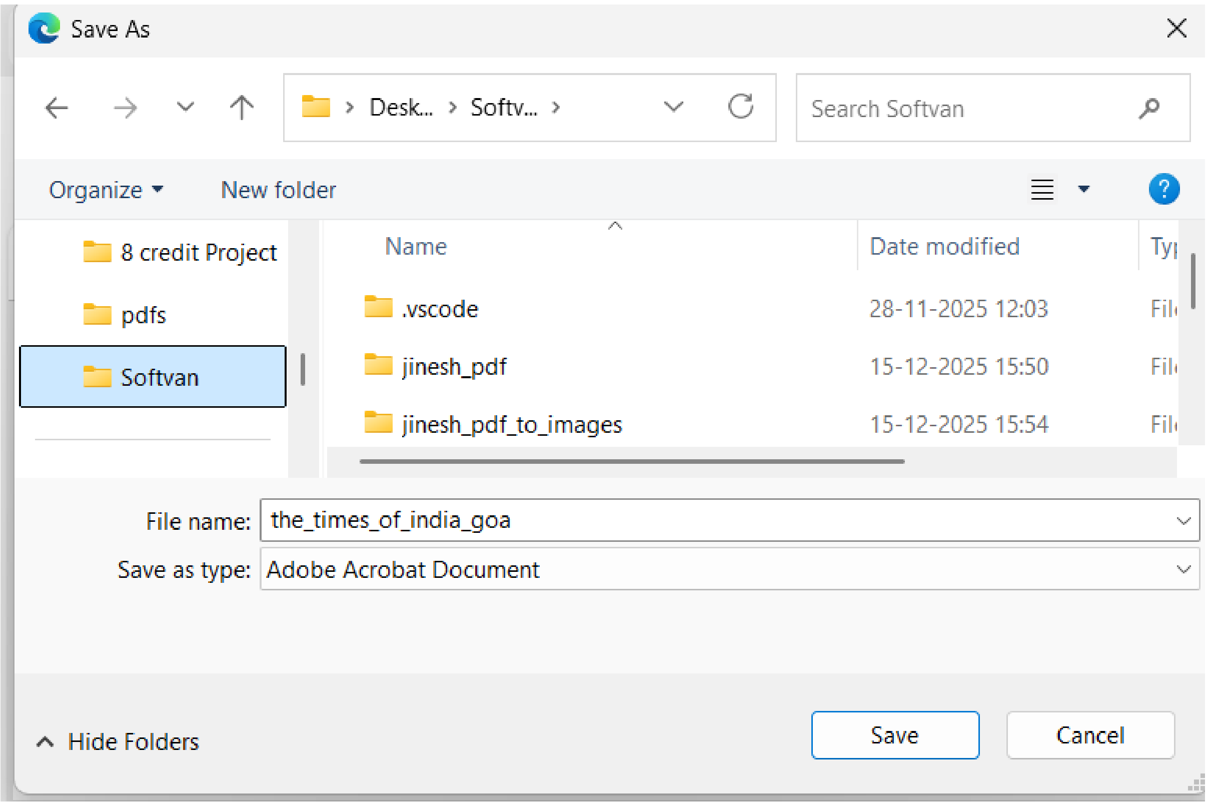
➢ Assign pdf



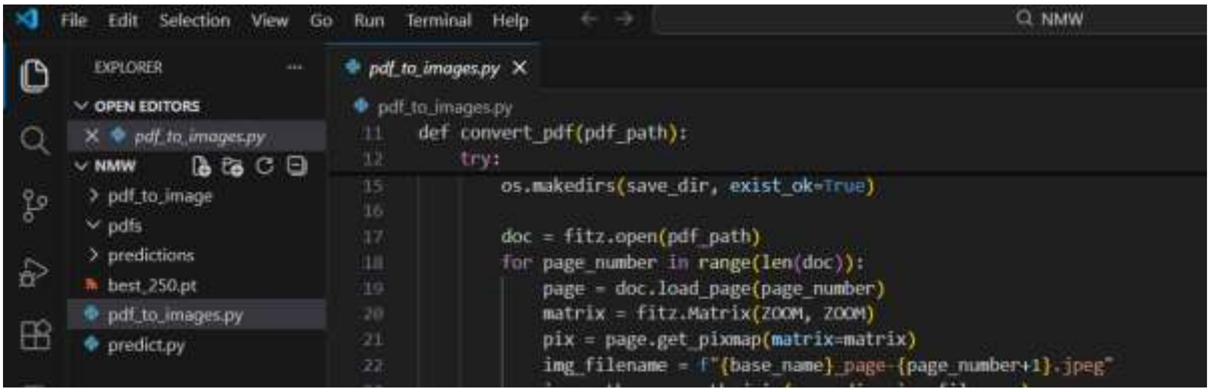
➢ Create a folder



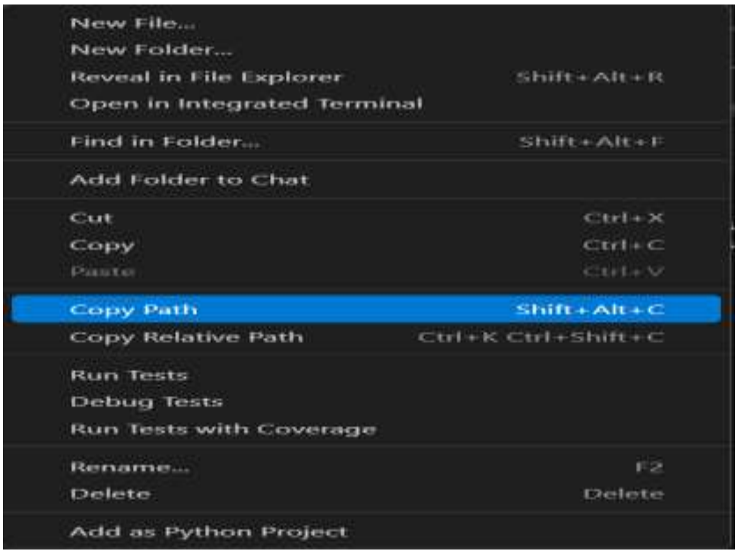
➢ Save pdf

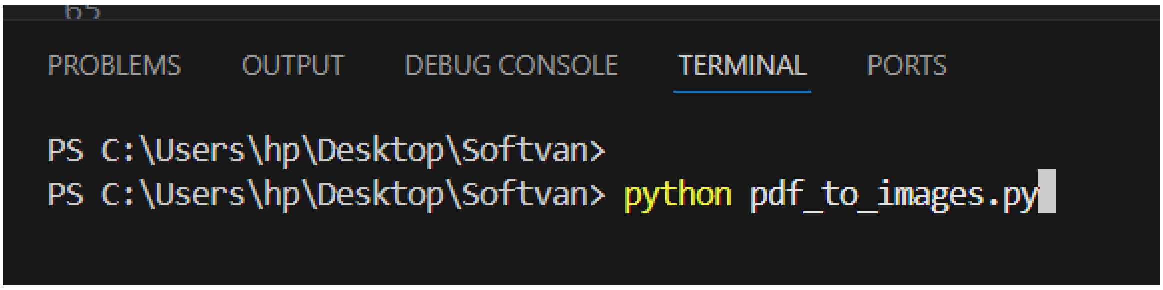


➢ Open folder VS Code



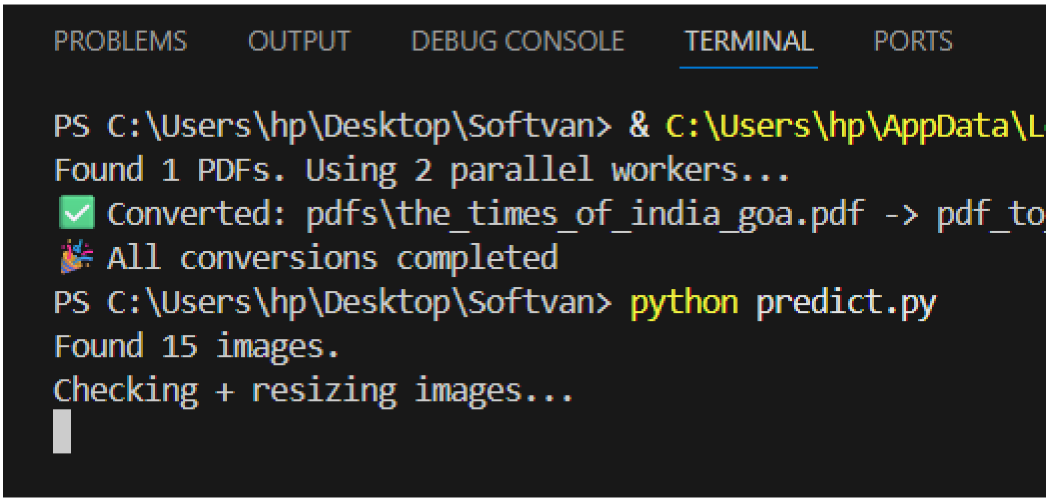
- **Open terminal paste python pdf_to_image.py**





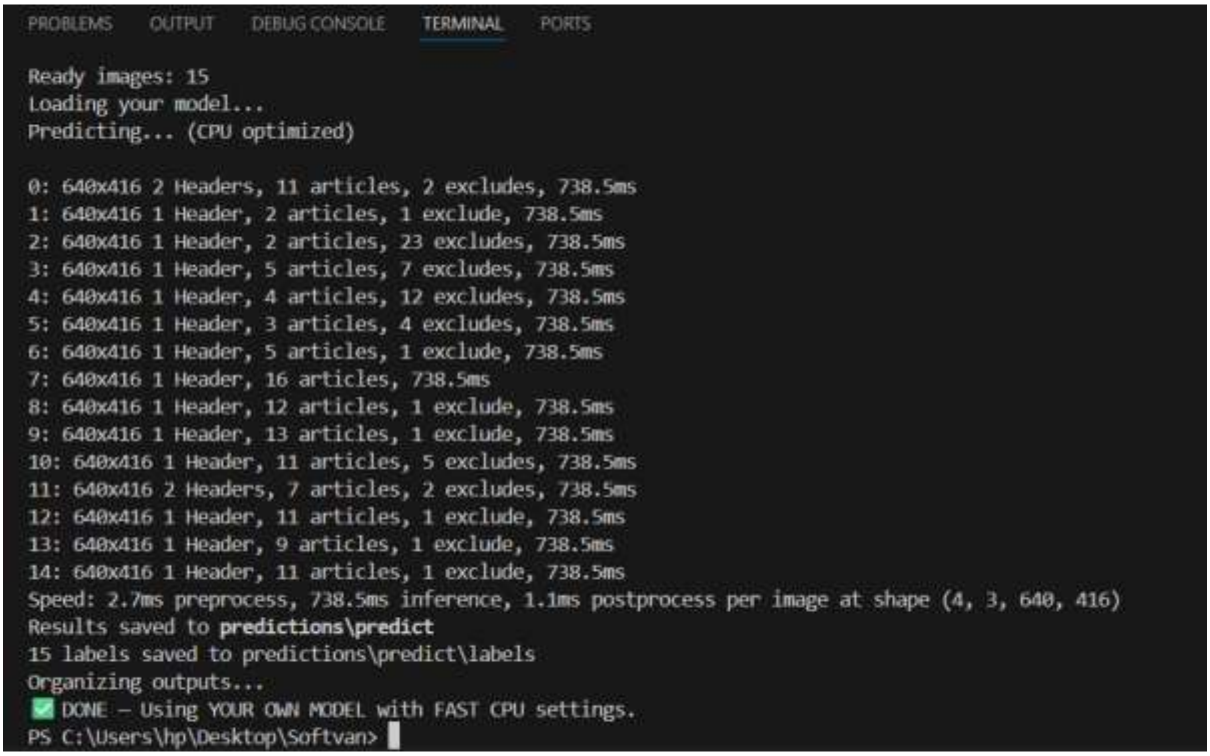
- **Convert the pdf to image**

- **Paste the predict.py**



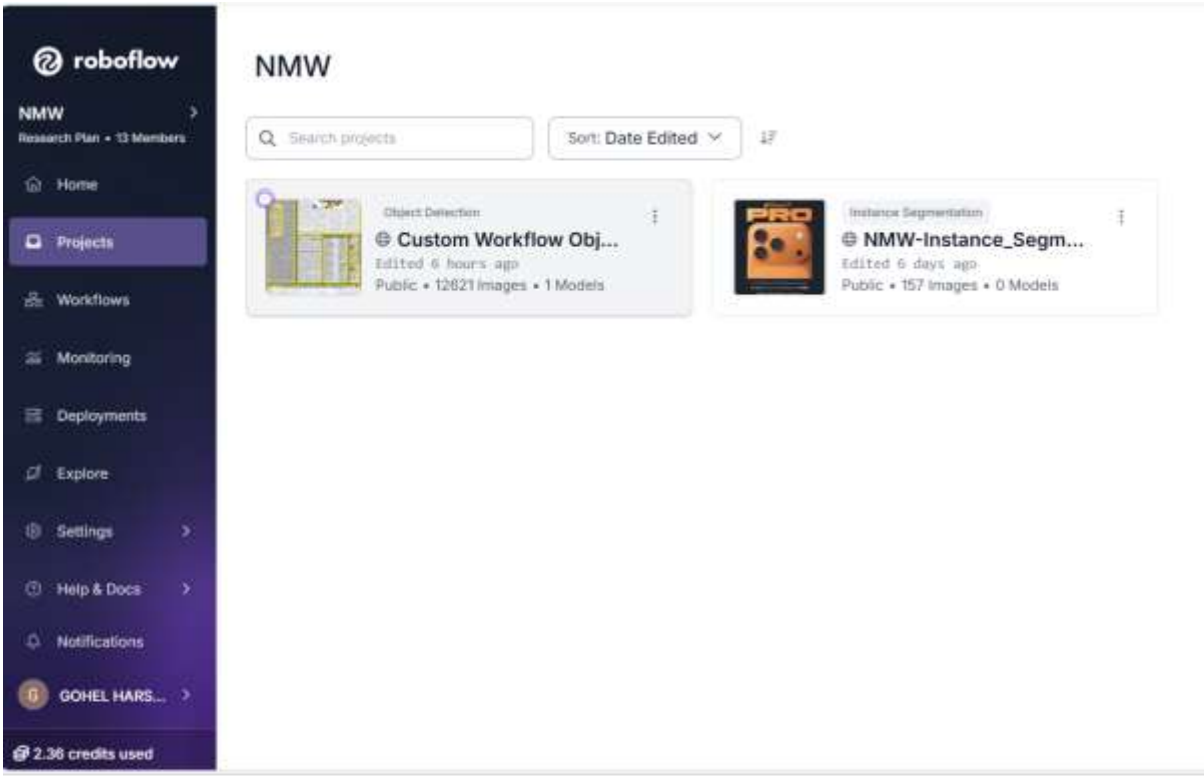
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\hp\Desktop\Softvan> & C:\Users\hp\AppData\L
Found 1 PDFs. Using 2 parallel workers...
✅ Converted: pdfs\the_times_of_india_goa.pdf -> pdf_to
🎉 All conversions completed
PS C:\Users\hp\Desktop\Softvan> python predict.py
Found 15 images.
Checking + resizing images...
```

➢ **Predict the image**

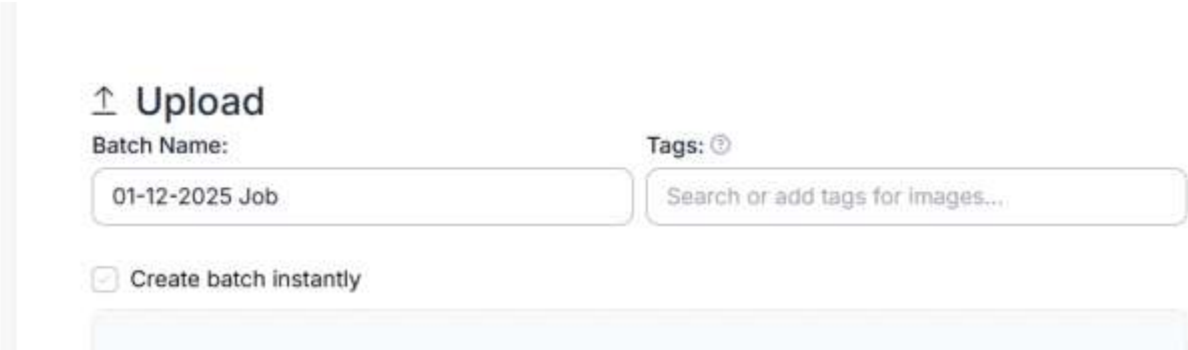


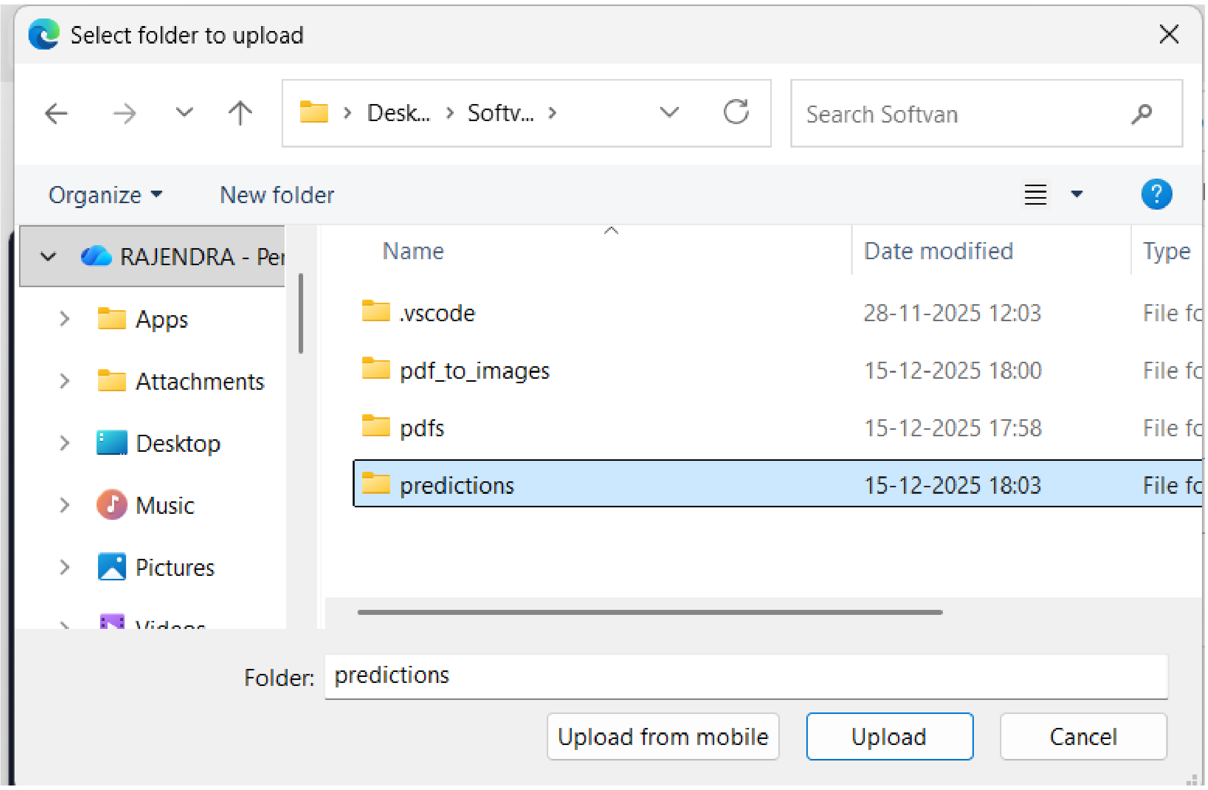
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

Ready images: 15
Loading your model...
Predicting... (CPU optimized)

0: 640x416 2 Headers, 11 articles, 2 excludes, 738.5ms
1: 640x416 1 Header, 2 articles, 1 exclude, 738.5ms
2: 640x416 1 Header, 2 articles, 23 excludes, 738.5ms
3: 640x416 1 Header, 5 articles, 7 excludes, 738.5ms
4: 640x416 1 Header, 4 articles, 12 excludes, 738.5ms
5: 640x416 1 Header, 3 articles, 4 excludes, 738.5ms
6: 640x416 1 Header, 5 articles, 1 exclude, 738.5ms
7: 640x416 1 Header, 16 articles, 738.5ms
8: 640x416 1 Header, 12 articles, 1 exclude, 738.5ms
9: 640x416 1 Header, 13 articles, 1 exclude, 738.5ms
10: 640x416 1 Header, 11 articles, 5 excludes, 738.5ms
11: 640x416 2 Headers, 7 articles, 2 excludes, 738.5ms
12: 640x416 1 Header, 11 articles, 1 exclude, 738.5ms
13: 640x416 1 Header, 9 articles, 1 exclude, 738.5ms
14: 640x416 1 Header, 11 articles, 1 exclude, 738.5ms
Speed: 2.7ms preprocess, 738.5ms inference, 1.1ms postprocess per image at shape (4, 3, 640, 416)
Results saved to predictions\predict
15 labels saved to predictions\predict\labels
Organizing outputs...
✅ DONE – Using YOUR OWN MODEL with FAST CPU settings.
PS C:\Users\hp\Desktop\Softvan>
```

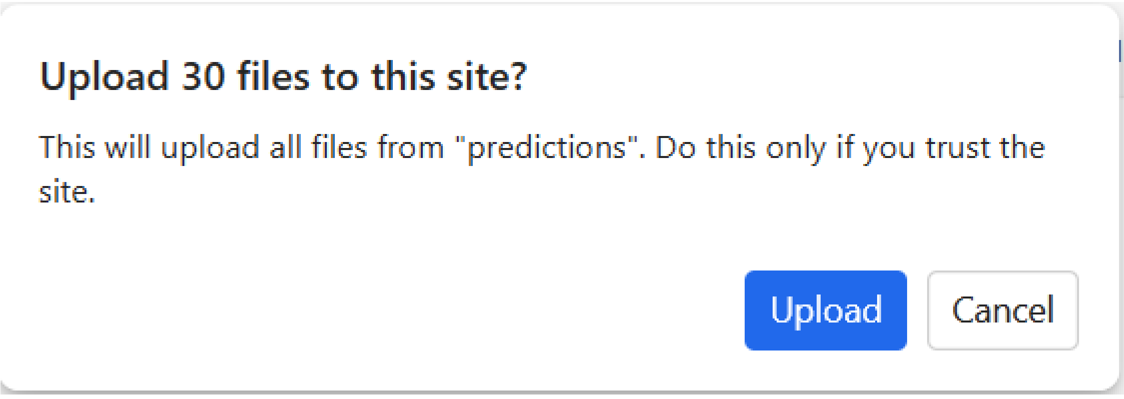
➢ **Open Roboflow**



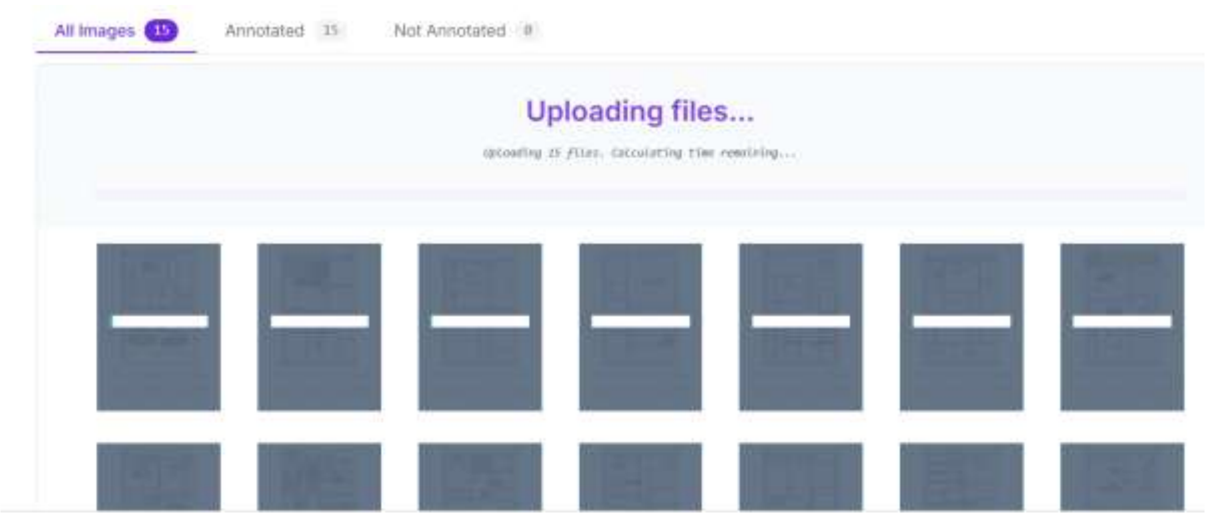
➢ **Upload your File/folder**



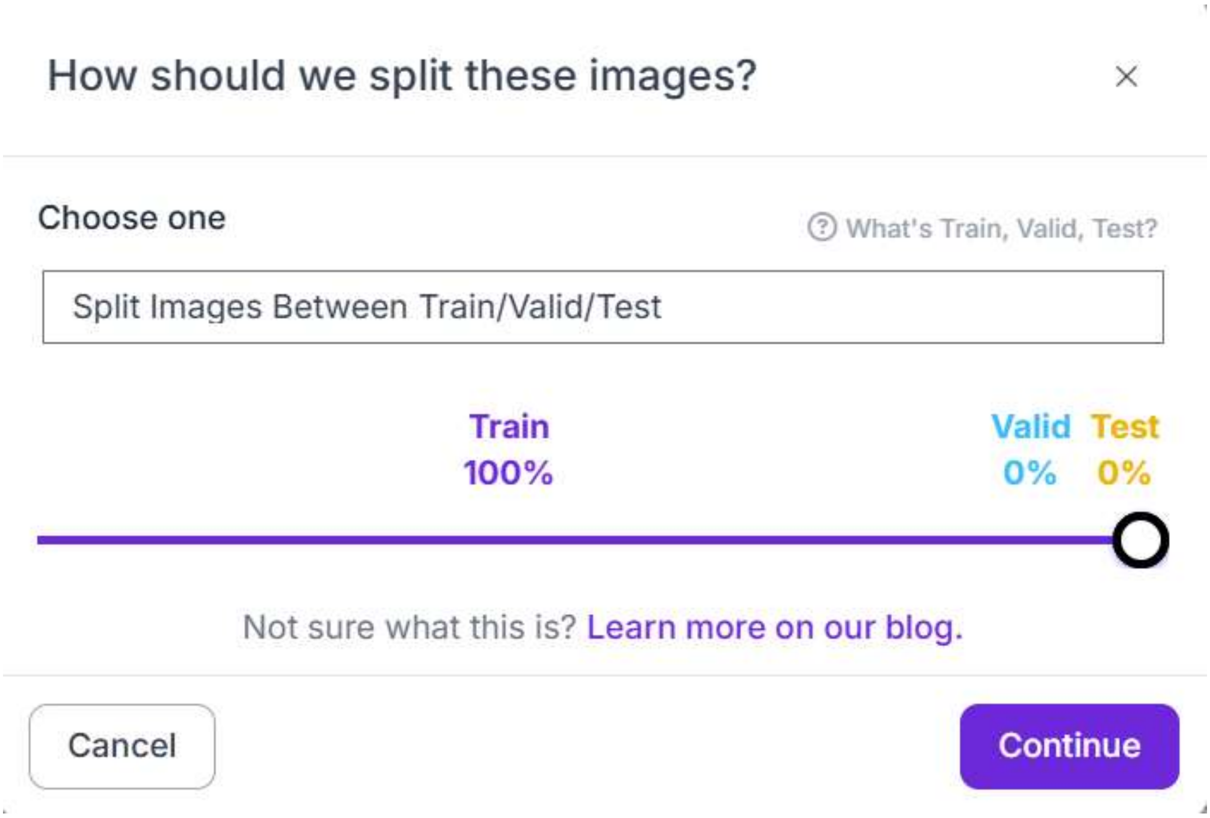




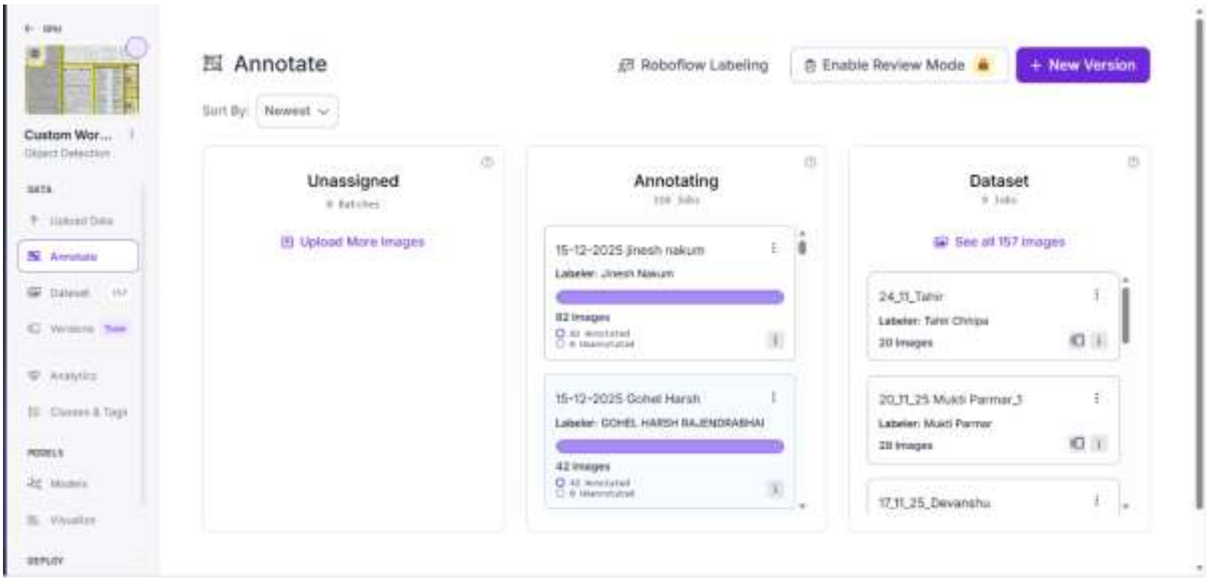
- **Uploading folder**



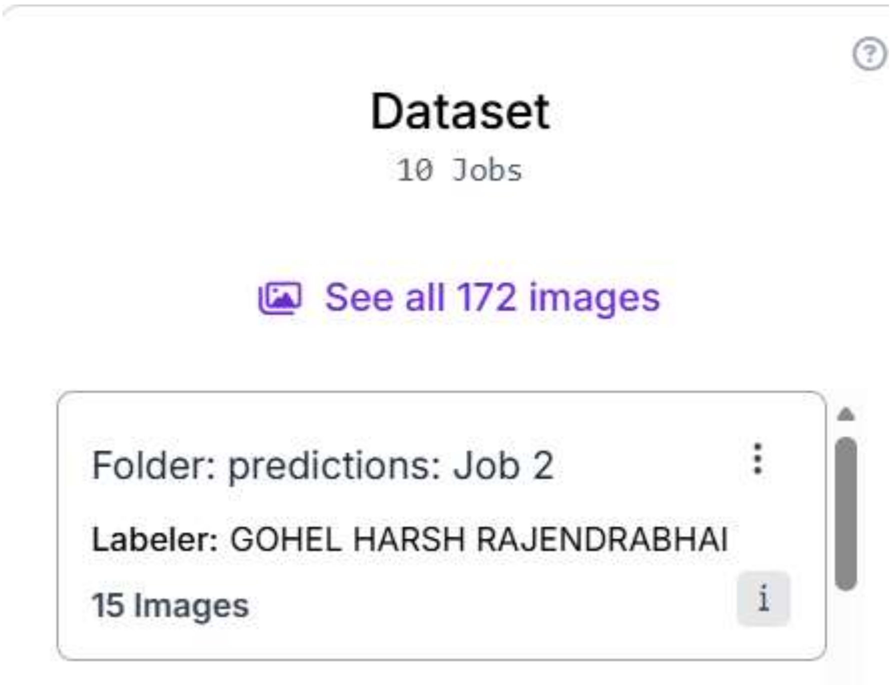
- **Train data**
-



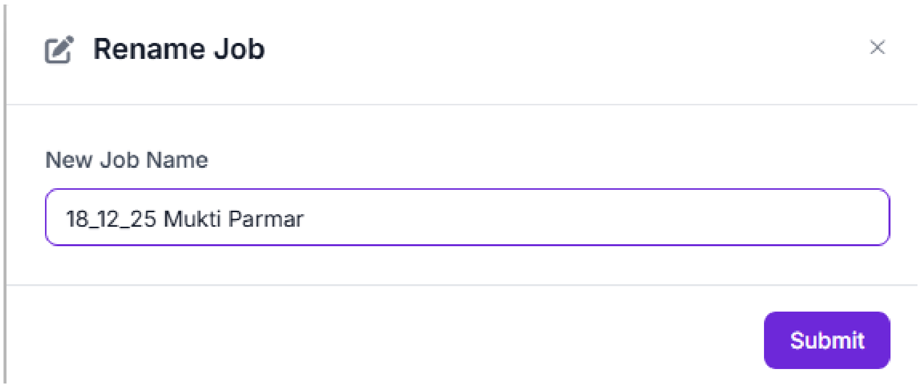
- **Click Annoted**



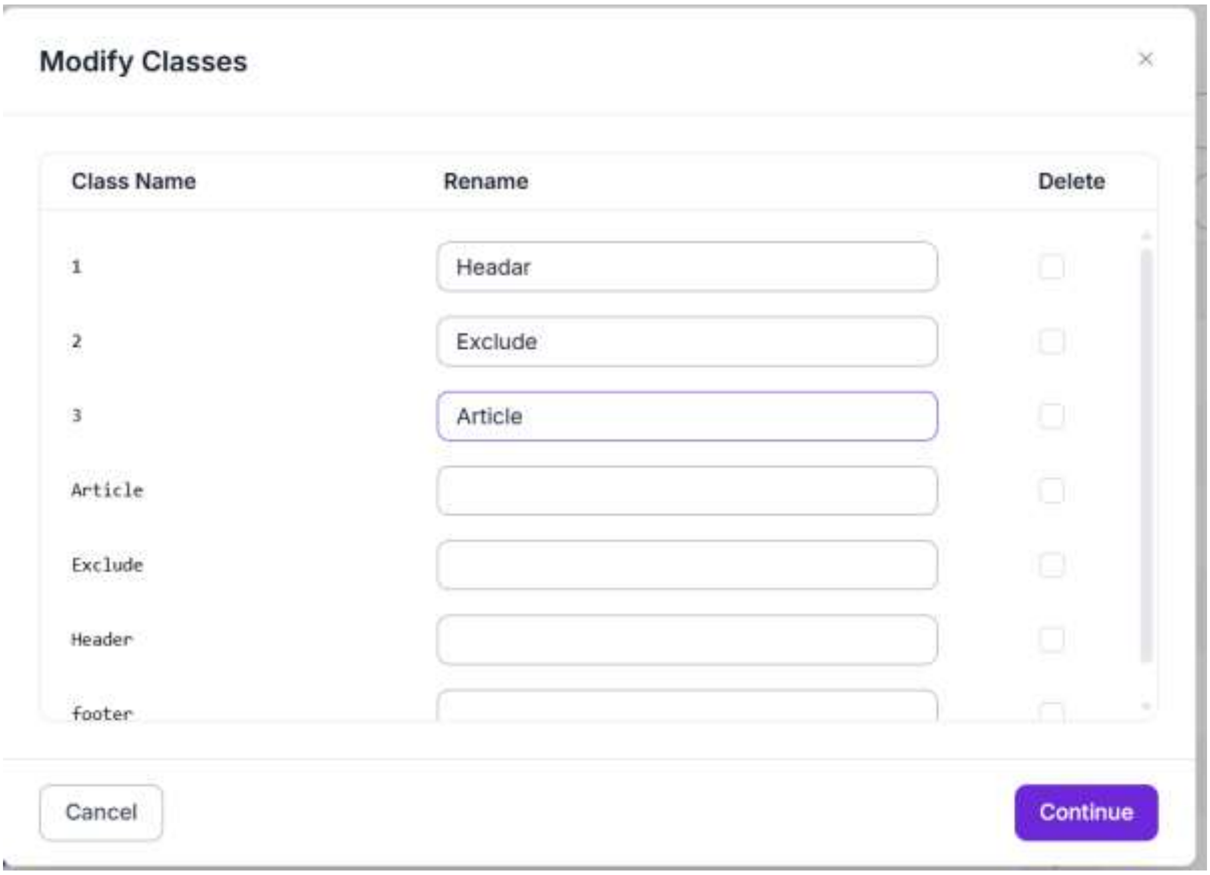
➢ Data in dataset



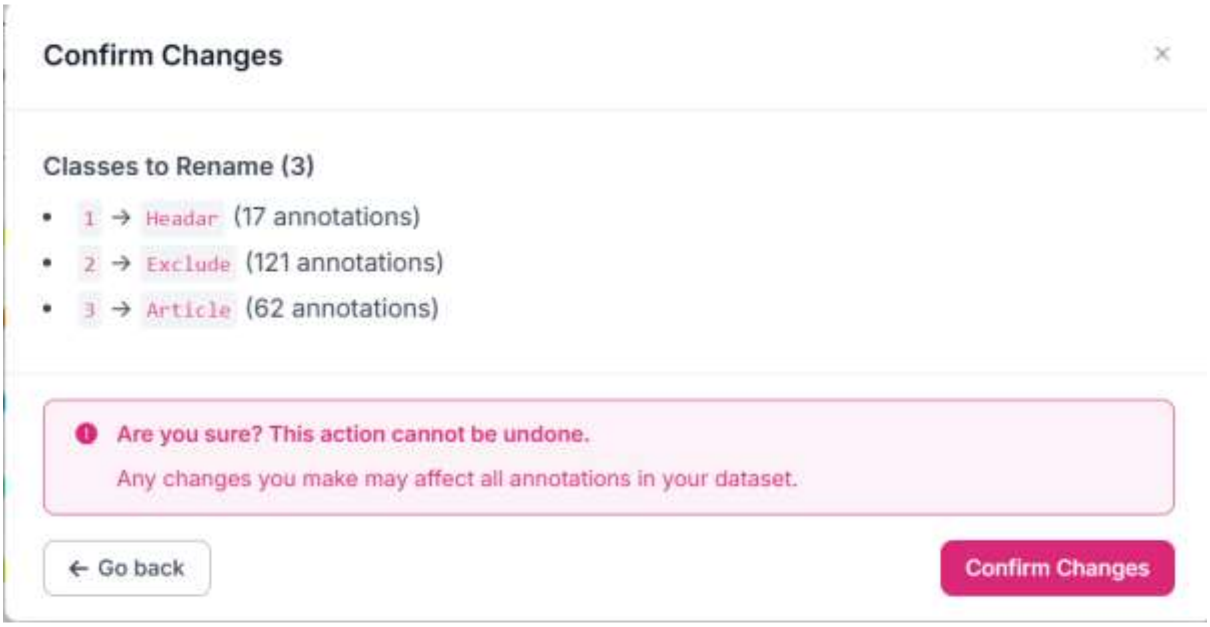
➢ Rename folder



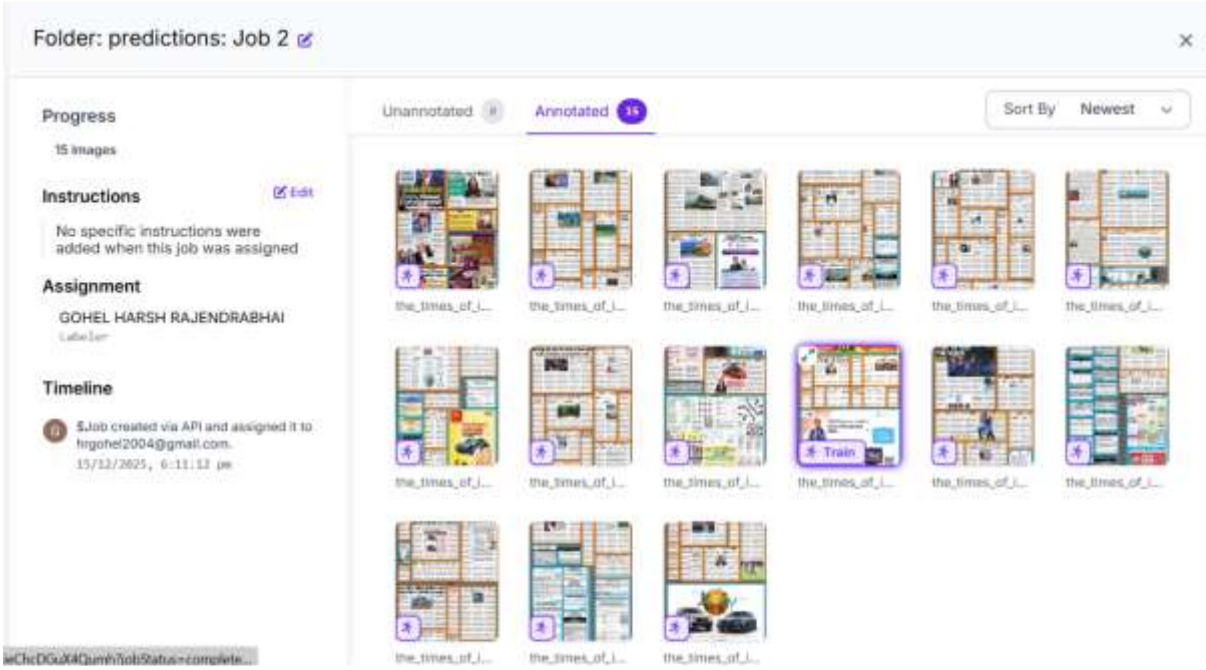
- **Modify the class name**



- **Change the class**



- **Starting validation**



➢ **Start the Annoted**

4. Tools and Technologies Used

4.1 Roboflow

- Image/PDF import
- Manual annotation
- Model-assisted labeling
- Version control
- Dataset export formats

4.2 PDF Handling Tools

- Adobe Acrobat
- Online PDF converters
- Python libraries (PyMuPDF, PDF2Image)

4.3 Internal Annotation Tools

- Company-specific labeling systems
- Custom annotation dashboards

4.4 Python Basics

I used Python scripts for:

- File renaming
- Folder automation
- Converting PDFs to images

4.5 Cloud Storage Systems

- Google Drive
- OneDrive
- Dropbox

Used for dataset uploading and sharing.

5. Skills Learned

5.1 Technical Skills

- End-to-end data annotation workflow
- Understanding ML dataset requirements
- Document classification
- Bounding box techniques
- Dataset versioning
- File management and PDF processing
- Roboflow usage for AI dataset creation
- Data cleaning and verification
- Introduction to automation using Python

5.2 Soft Skills

- Time management during repetitive tasks
- Attention to minute details
- Team communication with supervisors
- Following standard operating procedures
- Error identification and correction
- Task reporting and documentation

6. Challenges Faced (In Detail)

1. Low-quality PDFs
   Some documents had unclear text, blurred images, or poor scans.
   This made annotation difficult and time-consuming.
2. Strict Annotation Guidelines
   Labels needed to be extremely precise.
   Even small mistakes could affect ML model training.
3. Large Volume of Data
   Handling hundreds of pages required consistency and patience.
4. Learning New Annotation Tools
   At first, using Roboflow and company tools required practice.
5. Verification Workload
   Verification often took longer than annotation because every label had to be inspected.
6. Time Management
   Balancing speed and accuracy was challenging but improved over time.

7. Conclusion

This industry apprenticeship provided me with practical exposure to real-world AI/ML data annotation workflows. I gained hands-on experience in dataset preparation, manual annotation, verification, and the use of professional tools such as Roboflow.

The internship helped me understand the importance of data quality in machine learning systems and improved my technical accuracy, discipline, and professional approach. Overall, this experience has strengthened my foundation in AI/ML and will be beneficial for my future academic and professional growth.